Dynamic Programming — Complete DSA Sheet

All DP patterns covered | ~80 problems | Easy to Hard | LeetCode + GFG

1. DP FUNDAMENTALS — Learn These First

■ Understand memoization (top-down) vs tabulation (bottom-up) before solving

Concept Problems

■ Fibonacci — Recursion → Memo → Tabulation → Space Optimized
■ Climbing Stairs
■ Min Cost Climbing Stairs
■ Nth Tribonacci Number

2. 1D DP (8 problems)

■ dp[i] depends on dp[i-1] or dp[i-2] — simplest form of DP

Linear DP

■ House Robber I
■ House Robber II (circular)
■ Delete and Earn
■ Jump Game I — Can Reach?
■ Jump Game II — Min Jumps
■ Decode Ways
■ Perfect Squares
■ Integer Break

3. KNAPSACK PATTERN (10 problems)

■ 0/1 Knapsack is the FATHER of half the DP problems — master this first

0/1 Knapsack

■ 0/1 Knapsack — Classic GFG
■ Subset Sum Problem
■ Partition Equal Subset Sum
■ Target Sum
■ Last Stone Weight II

Unbounded Knapsack

■ Unbounded Knapsack — Classic GFG
■ Coin Change I — Min Coins
■ Coin Change II — Number of Ways
■ Rod Cutting Problem
■ Minimum Cost For Tickets

4. STRING DP (10 problems)

■ 2D dp[i][j] where i = index in string1, j = index in string2

Classic String DP

■ Longest Common Subsequence — LCS
■ Longest Common Substring
■ Edit Distance (Hard) ■
■ Delete Operation for Two Strings
■ Shortest Common Supersequence (Hard) ■
■ Longest Palindromic Subsequence
■ Longest Palindromic Substring
■ Palindrome Partitioning II — Min Cuts (Hard) ■
■ Word Break I
■ Word Break II (Hard) ■

5. SUBSEQUENCE DP (6 problems)

■ LIS family — very common in interviews

LIS Pattern

- ■ Longest Increasing Subsequence — LIS
- ■ Number of LIS
- ■ Russian Doll Envelopes (Hard) ■
- ■ Largest Divisible Subset
- ■ Maximum Length of Pair Chain
- ■ Wiggle Subsequence

6. GRID / 2D DP (8 problems)

■ dp[i][j] = answer for cell (i,j) — move right or down

Grid DP

- ■ Unique Paths I
- ■ Unique Paths II — with Obstacles
- ■ Minimum Path Sum
- ■ Triangle — Min Path Top to Bottom
- ■ Dungeon Game (Hard) ■
- ■ Cherry Pickup (Hard) ■
- ■ Maximal Square
- ■ Count Square Submatrices with All Ones

7. STOCK BUY/SELL DP (6 problems)

■ Classic state machine DP — hold / not-hold states

Stock Problems

- ■ Best Time to Buy and Sell Stock I — one transaction
- ■ Best Time to Buy and Sell Stock II — unlimited
- ■ Best Time to Buy and Sell Stock III — at most 2 (Hard) ■
- ■ Best Time to Buy and Sell Stock IV — at most K (Hard) ■
- ■ Best Time to Buy and Sell with Cooldown
- ■ Best Time to Buy and Sell with Transaction Fee

8. INTERVAL DP (5 problems)

■ dp[i][j] = answer for subarray from i to j — tricky, do after basics

Interval DP

- ■ Burst Balloons (Hard) ■
- ■ Matrix Chain Multiplication (Hard) ■
- ■ Minimum Cost to Merge Stones (Hard) ■
- ■ Strange Printer (Hard) ■
- ■ Palindrome Partitioning II

9. TREE DP (6 problems)

■ DP on tree — answer for subtree, combine children results

Tree DP

- ■ Diameter of Binary Tree
- ■ Binary Tree Maximum Path Sum (Hard) ■
- ■ House Robber III — on Tree
- ■ Longest Univalue Path
- ■ Maximum Product of Splitted Binary Tree
- ■ Count Good Nodes in Binary Tree

10. BITMASK DP (4 problems)

■ Use bitmask to represent subset state — for small N (N <= 20)

Bitmask DP

- ■ Traveling Salesman Problem — TSP (Hard) ■
- ■ Minimum Cost to Visit Every Node in Graph (Hard) ■
- ■ Partition to K Equal Sum Subsets
- ■ Assign Cookies / Maximize Score After N Operations

11. DP ON GRAPHS (4 problems)

■ Shortest path with DP — Bellman Ford family

Graph DP

- ■ Cheapest Flights Within K Stops — Bellman Ford
- ■ Number of Ways to Reach Destination
- ■ Longest Path in DAG
- ■ Count All Paths from Source to Destination

12. MISCELLANEOUS DP (6 problems)

■ Important problems that don't fit one pattern — still very common

Mixed

- ■ Egg Drop Problem (Hard) ■
- ■ Stone Game
- ■ Predict the Winner
- ■ Wildcard Matching (Hard) ■
- ■ Regular Expression Matching (Hard) ■
- ■ Arithmetic Slices II

SUMMARY

#	DP Pattern	Problems	Hard
1	Fundamentals	4	0
2	1D DP	8	0
3	Knapsack (0/1 + Unbounded)	10	0
4	String DP	10	4
5	Subsequence / LIS	6	1
6	Grid / 2D DP	8	2
7	Stock Buy/Sell	6	2
8	Interval DP	5	4
9	Tree DP	6	1
10	Bitmask DP	4	2
11	DP on Graphs	4	0
12	Miscellaneous DP	6	3
TOTAL		**77**	**19**

Recommended Order: Fundamentals → 1D DP → Knapsack → String DP → Subsequence → Grid DP → Stocks → Interval DP → Tree DP → Bitmask → Graph DP

■ Red = Hard | ■ = Not done | Knapsack is the most important pattern — spend 3-4 days on it